Bayesian Regression

```r
library(geoR)
```

```
## Warning in fun(libname, pkgname): couldn't connect to display ":0"
## ----------------------------------------------------------------
##  Analysis of Geostatistical Data
##  For an Introduction to geoR go to http://www.leg.ufpr.br/geoR
##  geoR version 1.9-6 (built on 2025-08-29) is now loaded
## ----------------------------------------------------------------
```

```r
library(mvnfast)
library(MCMCpack)
```

```
## Loading required package: coda
## Loading required package: MASS
## ##
## ## Markov Chain Monte Carlo Package (MCMCpack)
## ## Copyright (C) 2003-2026 Andrew D. Martin, Kevin M. Quinn, and Jong Hee Park
## ##
## ## Support provided by the U.S. National Science Foundation
## ## (Grants SES-0350646 and SES-0350613)
## ##
```

```r
library(brms)
```

```
## Loading required package: Rcpp
## Loading 'brms' package (version 2.23.0). Useful instructions
## can be found by typing help('brms'). A more detailed introduction
## to the package is available through vignette('brms_overview').
##
## Attaching package: 'brms'
## The following objects are masked from 'package:MCMCpack':
##
##     ddirichlet, rdirichlet
## The following object is masked from 'package:stats':
##
##     ar
```

loading data

```r
X_train <- read.csv("train-test/X_train_diff.csv")
X_test <- read.csv("train-test/X_test_diff.csv")
y_train <- read.csv("train-test/y_train.csv")
y_test <- read.csv("train-test/y_test.csv")
```

```r
train_df <- cbind(X_train, POINT_DIFF = y_train$POINT_DIFF)
test_df <- cbind(X_test, POINT_DIFF = y_test$POINT_DIFF)

cat("Training rows:", nrow(train_df), "\n")
```

```
## Training rows: 6822
```

```r
cat("Test rows:", nrow(test_df),  "\n")
```

```
## Test rows: 870
```

fitting model

```r
# note the "." means all other columns

prior <- default_prior(POINT_DIFF ~ .,
                       data = train_df,
                       family = gaussian())

manual_prior <- c(
  prior(student_t(3, 0, 5), class = "Intercept"),
  prior(student_t(3, 0, 5), class = "sigma"),
  prior(normal(0, 1), class = "b")
)

fit <- brm(POINT_DIFF ~ .,
           family = gaussian(),
           data = train_df,
           chains = 4,
           iter = 1000,
           warmup = 180,
           thin = 2,
           prior = manual_prior)
```

```
## Compiling Stan program...

## Trying to compile a simple C file

## Running /Library/Frameworks/R.framework/Resources/bin/R CMD SHLIB foo.c
## using C compiler: 'Apple clang version 17.0.0 (clang-1700.0.13.5)'
## using SDK: 'MacOSX15.5.sdk'
## clang -arch arm64 -std=gnu2x -I"/Library/Frameworks/R.framework/Resources/include" -DNDEBUG   -I"/Li
## In file included from <built-in>:1:
## In file included from /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/StanHeade
## In file included from /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/RcppEigen/
## In file included from /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/RcppEigen/
## /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/RcppEigen/include/Eigen/src/Cor
##   679 | #include <cmath>
##       |          ^~~~~~~
## 1 error generated.
## make: *** [foo.o] Error 1

## Start sampling

##
## SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 1).
```

```
## Chain 1:
## Chain 1: Gradient evaluation took 0.000191 seconds
## Chain 1: 1000 transitions using 10 leapfrog steps per transition would take 1.91 seconds.
## Chain 1: Adjust your expectations accordingly!
## Chain 1:
## Chain 1:
## Chain 1: Iteration:   1 / 1000 [  0%]  (Warmup)
## Chain 1: Iteration: 100 / 1000 [ 10%]  (Warmup)
## Chain 1: Iteration: 181 / 1000 [ 18%]  (Sampling)
## Chain 1: Iteration: 280 / 1000 [ 28%]  (Sampling)
## Chain 1: Iteration: 380 / 1000 [ 38%]  (Sampling)
## Chain 1: Iteration: 480 / 1000 [ 48%]  (Sampling)
## Chain 1: Iteration: 580 / 1000 [ 58%]  (Sampling)
## Chain 1: Iteration: 680 / 1000 [ 68%]  (Sampling)
## Chain 1: Iteration: 780 / 1000 [ 78%]  (Sampling)
## Chain 1: Iteration: 880 / 1000 [ 88%]  (Sampling)
## Chain 1: Iteration: 980 / 1000 [ 98%]  (Sampling)
## Chain 1: Iteration: 1000 / 1000 [100%]  (Sampling)
## Chain 1:
## Chain 1:  Elapsed Time: 4.727 seconds (Warm-up)
## Chain 1:                2.037 seconds (Sampling)
## Chain 1:                6.764 seconds (Total)
## Chain 1:
##
## SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 2).
## Chain 2:
## Chain 2: Gradient evaluation took 0.000154 seconds
## Chain 2: 1000 transitions using 10 leapfrog steps per transition would take 1.54 seconds.
## Chain 2: Adjust your expectations accordingly!
## Chain 2:
## Chain 2:
## Chain 2: Iteration:   1 / 1000 [  0%]  (Warmup)
## Chain 2: Iteration: 100 / 1000 [ 10%]  (Warmup)
## Chain 2: Iteration: 181 / 1000 [ 18%]  (Sampling)
## Chain 2: Iteration: 280 / 1000 [ 28%]  (Sampling)
## Chain 2: Iteration: 380 / 1000 [ 38%]  (Sampling)
## Chain 2: Iteration: 480 / 1000 [ 48%]  (Sampling)
## Chain 2: Iteration: 580 / 1000 [ 58%]  (Sampling)
## Chain 2: Iteration: 680 / 1000 [ 68%]  (Sampling)
## Chain 2: Iteration: 780 / 1000 [ 78%]  (Sampling)
## Chain 2: Iteration: 880 / 1000 [ 88%]  (Sampling)
## Chain 2: Iteration: 980 / 1000 [ 98%]  (Sampling)
## Chain 2: Iteration: 1000 / 1000 [100%]  (Sampling)
## Chain 2:
## Chain 2:  Elapsed Time: 4.227 seconds (Warm-up)
## Chain 2:                2.059 seconds (Sampling)
## Chain 2:                6.286 seconds (Total)
## Chain 2:
##
## SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 3).
## Chain 3:
## Chain 3: Gradient evaluation took 0.000166 seconds
## Chain 3: 1000 transitions using 10 leapfrog steps per transition would take 1.66 seconds.
## Chain 3: Adjust your expectations accordingly!
```

```
## Chain 3:
## Chain 3:
## Chain 3: Iteration:   1 / 1000 [  0%]  (Warmup)
## Chain 3: Iteration: 100 / 1000 [ 10%]  (Warmup)
## Chain 3: Iteration: 181 / 1000 [ 18%]  (Sampling)
## Chain 3: Iteration: 280 / 1000 [ 28%]  (Sampling)
## Chain 3: Iteration: 380 / 1000 [ 38%]  (Sampling)
## Chain 3: Iteration: 480 / 1000 [ 48%]  (Sampling)
## Chain 3: Iteration: 580 / 1000 [ 58%]  (Sampling)
## Chain 3: Iteration: 680 / 1000 [ 68%]  (Sampling)
## Chain 3: Iteration: 780 / 1000 [ 78%]  (Sampling)
## Chain 3: Iteration: 880 / 1000 [ 88%]  (Sampling)
## Chain 3: Iteration: 980 / 1000 [ 98%]  (Sampling)
## Chain 3: Iteration: 1000 / 1000 [100%]  (Sampling)
## Chain 3:
## Chain 3:  Elapsed Time: 4.809 seconds (Warm-up)
## Chain 3:                2.028 seconds (Sampling)
## Chain 3:                6.837 seconds (Total)
## Chain 3:
##
## SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 4).
## Chain 4:
## Chain 4: Gradient evaluation took 0.000153 seconds
## Chain 4: 1000 transitions using 10 leapfrog steps per transition would take 1.53 seconds.
## Chain 4: Adjust your expectations accordingly!
## Chain 4:
## Chain 4:
## Chain 4: Iteration:   1 / 1000 [  0%]  (Warmup)
## Chain 4: Iteration: 100 / 1000 [ 10%]  (Warmup)
## Chain 4: Iteration: 181 / 1000 [ 18%]  (Sampling)
## Chain 4: Iteration: 280 / 1000 [ 28%]  (Sampling)
## Chain 4: Iteration: 380 / 1000 [ 38%]  (Sampling)
## Chain 4: Iteration: 480 / 1000 [ 48%]  (Sampling)
## Chain 4: Iteration: 580 / 1000 [ 58%]  (Sampling)
## Chain 4: Iteration: 680 / 1000 [ 68%]  (Sampling)
## Chain 4: Iteration: 780 / 1000 [ 78%]  (Sampling)
## Chain 4: Iteration: 880 / 1000 [ 88%]  (Sampling)
## Chain 4: Iteration: 980 / 1000 [ 98%]  (Sampling)
## Chain 4: Iteration: 1000 / 1000 [100%]  (Sampling)
## Chain 4:
## Chain 4:  Elapsed Time: 4.834 seconds (Warm-up)
## Chain 4:                2.045 seconds (Sampling)
## Chain 4:                6.879 seconds (Total)
## Chain 4:
```

```r
summary(fit)
```

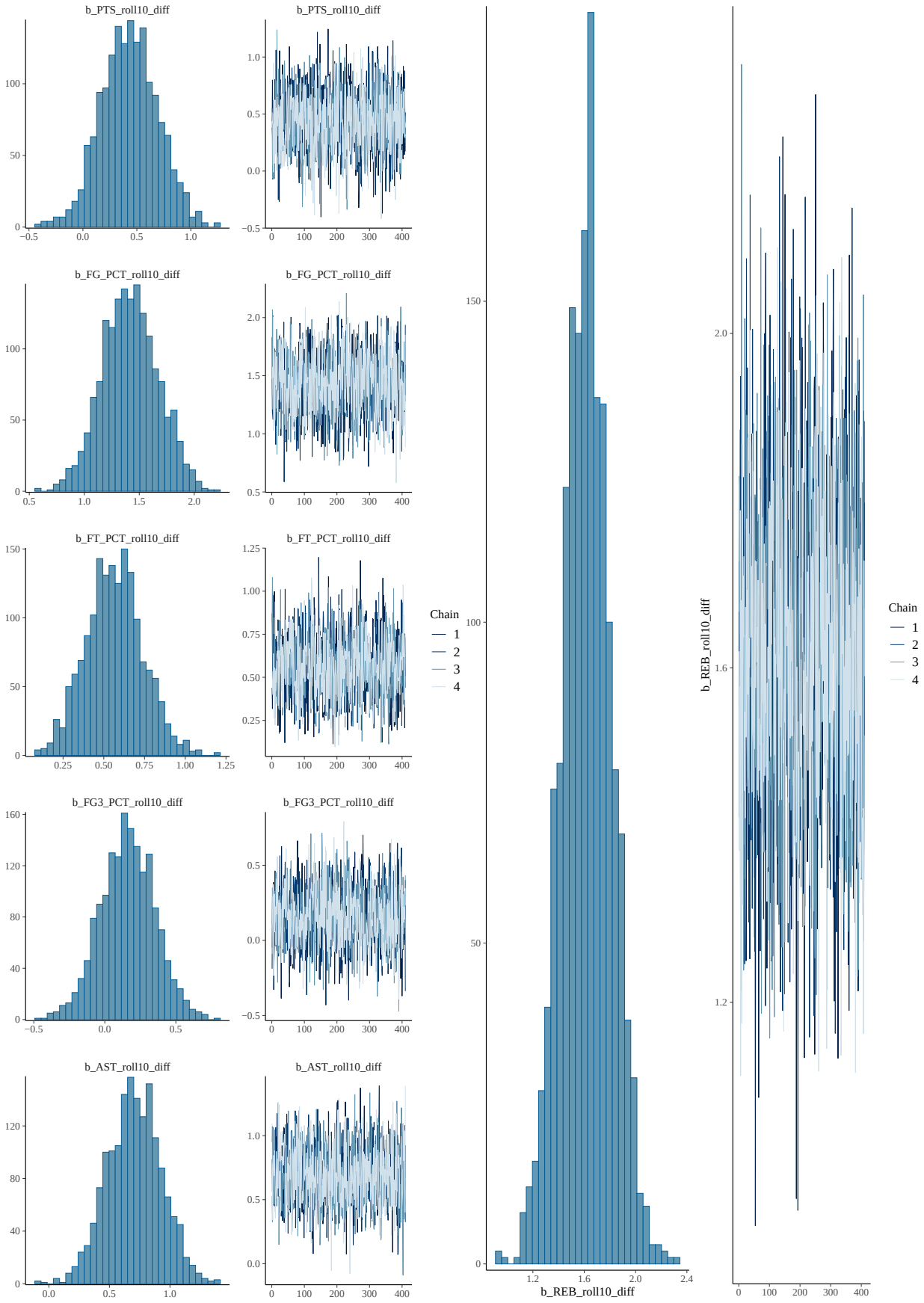
```
##  Family: gaussian
##   Links: mu = identity
## Formula: POINT_DIFF ~ PTS_roll10_diff + FG_PCT_roll10_diff + FT_PCT_roll10_diff + FG3_PCT_roll10_dif
##    Data: train_df (Number of observations: 6822)
##   Draws: 4 chains, each with iter = 1000; warmup = 180; thin = 2;
##          total post-warmup draws = 1640
##
## Regression Coefficients:
```

```
##                      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## Intercept                2.67      0.30     2.09     3.24 1.00     1453     1329
## PTS_roll10_diff          0.43      0.26    -0.07     0.94 1.00     1347     1223
## FG_PCT_roll10_diff       1.42      0.25     0.93     1.92 1.00     1379     1500
## FT_PCT_roll10_diff       0.57      0.17     0.23     0.91 1.00     1267     1225
## FG3_PCT_roll10_diff      0.16      0.19    -0.22     0.53 1.00     1114     1092
## AST_roll10_diff          0.71      0.22     0.26     1.12 1.00     1335     1489
## REB_roll10_diff          1.62      0.19     1.25     1.99 1.00     1366     1424
## HOME_Bucks               0.85      0.66    -0.40     2.14 1.00     1305     1264
## HOME_Bulls              -2.34      0.67    -3.64    -1.00 1.00     1402     1367
## HOME_Cavaliers          -0.46      0.67    -1.77     0.82 1.00     1198     1106
## HOME_Celtics             1.59      0.65     0.31     2.90 1.00     1293     1260
## HOME_Clippers            1.12      0.66    -0.22     2.39 1.00     1371     1345
## HOME_Grizzlies          -0.28      0.68    -1.58     1.08 1.00     1269     1372
## HOME_Hawks              -1.53      0.68    -2.85    -0.15 1.00     1245     1069
## HOME_Heat                0.94      0.67    -0.37     2.22 1.00     1126     1187
## HOME_Hornets             0.14      0.68    -1.17     1.50 1.00     1346     1314
## HOME_Jazz                1.98      0.68     0.66     3.35 1.00     1231     1282
## HOME_Kings              -1.99      0.68    -3.26    -0.63 1.00     1153     1158
## HOME_Knicks             -2.61      0.72    -4.02    -1.23 1.00     1194     1188
## HOME_Lakers             -1.25      0.66    -2.51     0.04 1.00     1465     1398
## HOME_Magic              -1.34      0.68    -2.65    -0.04 1.00     1350     1284
## HOME_Mavericks           0.31      0.69    -1.01     1.64 1.00     1163     1257
## HOME_Nets               -2.39      0.66    -3.62    -1.06 1.00     1399     1282
## HOME_Nuggets             0.54      0.67    -0.74     1.91 1.00     1486     1254
## HOME_Pacers              1.24      0.67    -0.06     2.56 1.00     1291     1273
## HOME_Pelicans           -1.76      0.69    -3.11    -0.40 1.00     1249     1400
## HOME_Pistons             0.46      0.69    -0.88     1.78 1.01     1413     1433
## HOME_Raptors             2.87      0.65     1.57     4.07 1.01     1234     1391
## HOME_Rockets             2.50      0.69     1.19     3.82 1.00     1257     1326
## HOME_Spurs               2.14      0.66     0.89     3.49 1.00     1325     1337
## HOME_Suns               -2.85      0.71    -4.27    -1.45 1.00      978     1203
## HOME_Thunder             1.77      0.69     0.38     3.10 1.00     1129     1013
## HOME_Timberwolves       -0.89      0.70    -2.28     0.53 1.00     1428     1159
## HOME_Trail.Blazers       0.69      0.67    -0.64     1.97 1.00     1337     1344
## HOME_Warriors            1.64      0.66     0.37     2.90 1.00     1382     1189
## HOME_Wizards            -0.17      0.72    -1.54     1.18 1.00     1073     1229
## AWAY_Bucks              -0.79      0.68    -2.05     0.54 1.00     1150     1031
## AWAY_Bulls               1.22      0.67    -0.07     2.50 1.01     1253     1186
## AWAY_Cavaliers          -0.04      0.63    -1.27     1.23 1.00     1338     1217
## AWAY_Celtics            -2.28      0.66    -3.54    -1.00 1.00     1327     1342
## AWAY_Clippers           -1.39      0.66    -2.62    -0.12 1.00     1270     1290
## AWAY_Grizzlies           0.28      0.69    -1.11     1.61 1.00     1273     1192
## AWAY_Hawks               1.27      0.68    -0.05     2.62 1.00     1605     1571
## AWAY_Heat               -0.65      0.66    -1.91     0.58 1.00     1024     1236
## AWAY_Hornets            -0.10      0.68    -1.40     1.28 1.00     1155     1116
## AWAY_Jazz               -2.16      0.69    -3.49    -0.83 1.00     1345     1215
## AWAY_Kings               0.82      0.70    -0.55     2.19 1.00     1200     1293
## AWAY_Knicks              1.89      0.70     0.46     3.22 1.00     1273     1340
## AWAY_Lakers              1.08      0.69    -0.27     2.41 1.00     1218     1125
## AWAY_Magic               0.63      0.67    -0.67     1.83 1.01     1298     1234
## AWAY_Mavericks          -0.07      0.69    -1.45     1.30 1.00     1377     1404
## AWAY_Nets                1.51      0.69     0.17     2.90 1.00     1174     1291
## AWAY_Nuggets             0.40      0.67    -0.92     1.69 1.00     1204     1371
```

```
## AWAY_Pacers            -0.44      0.67    -1.73     0.88 1.00     1192     1119
## AWAY_Pelicans           0.63      0.69    -0.70     2.03 1.00     1197     1372
## AWAY_Pistons            0.37      0.67    -0.92     1.63 1.00     1352     1435
## AWAY_Raptors           -2.67      0.67    -3.94    -1.31 1.00     1103     1023
## AWAY_Rockets           -2.87      0.69    -4.21    -1.55 1.00     1396     1434
## AWAY_Spurs             -1.20      0.65    -2.52     0.05 1.00     1274     1311
## AWAY_Suns               2.37      0.69     1.04     3.68 1.00     1252     1139
## AWAY_Thunder           -1.15      0.66    -2.45     0.12 1.00     1349     1269
## AWAY_Timberwolves       0.70      0.69    -0.69     2.02 1.00     1551     1150
## AWAY_Trail.Blazers      0.06      0.67    -1.21     1.40 1.00     1373     1213
## AWAY_Warriors          -1.71      0.70    -3.17    -0.40 1.00     1383     1375
## AWAY_Wizards            0.97      0.70    -0.37     2.38 1.00     1187     1172
##
## Further Distributional Parameters:
##       Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sigma    13.22      0.12    13.01    13.44 1.00     1452     1126
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

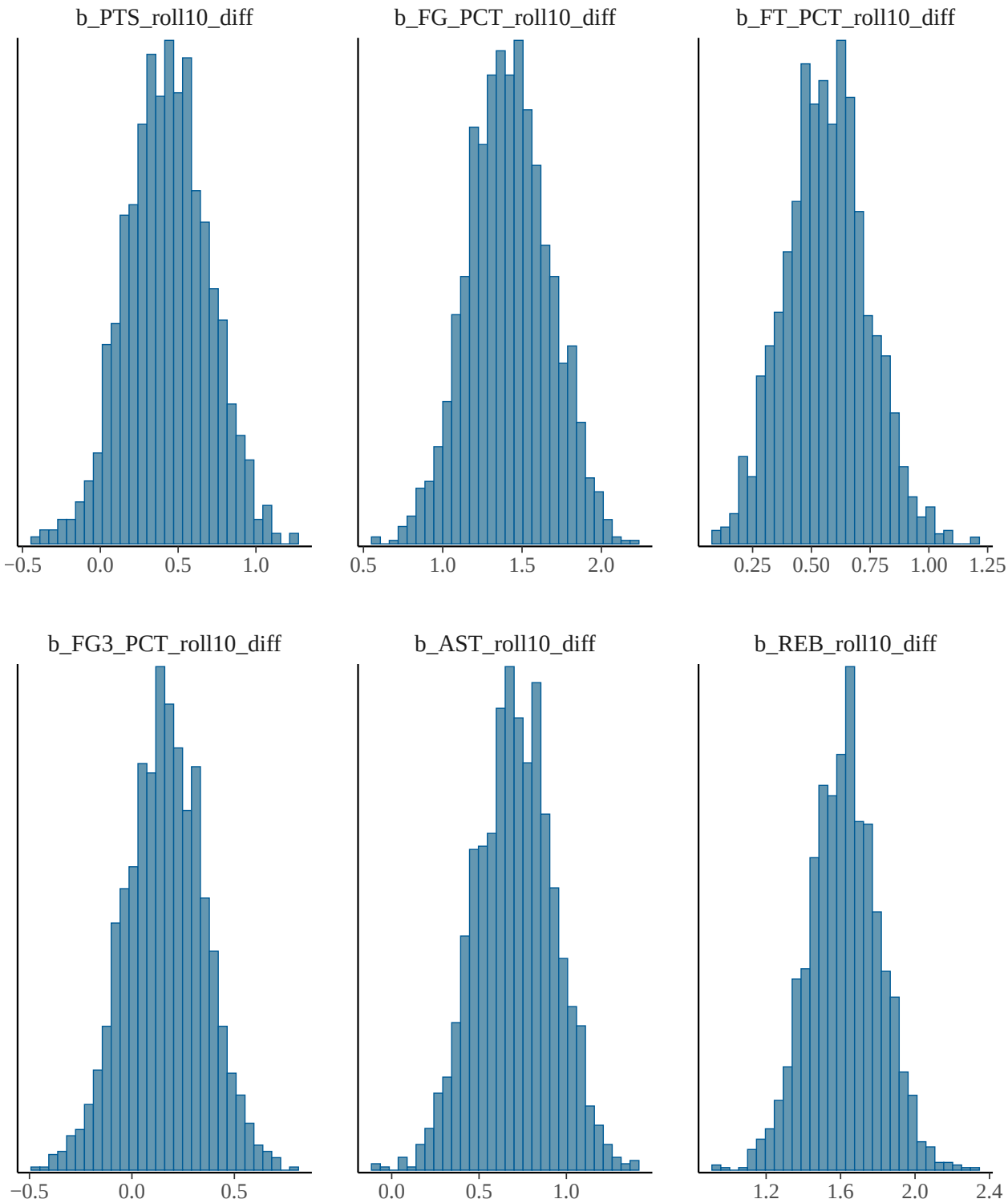
chain trace plots/marginal posteriors for the 12 rolling features

```r
rolling_vars <- c(
  "b_PTS_roll10_diff",
  "b_FG_PCT_roll10_diff",
  "b_FT_PCT_roll10_diff",
  "b_FG3_PCT_roll10_diff",
  "b_AST_roll10_diff",
  "b_REB_roll10_diff"
)

plot(fit, variable = rolling_vars)
```

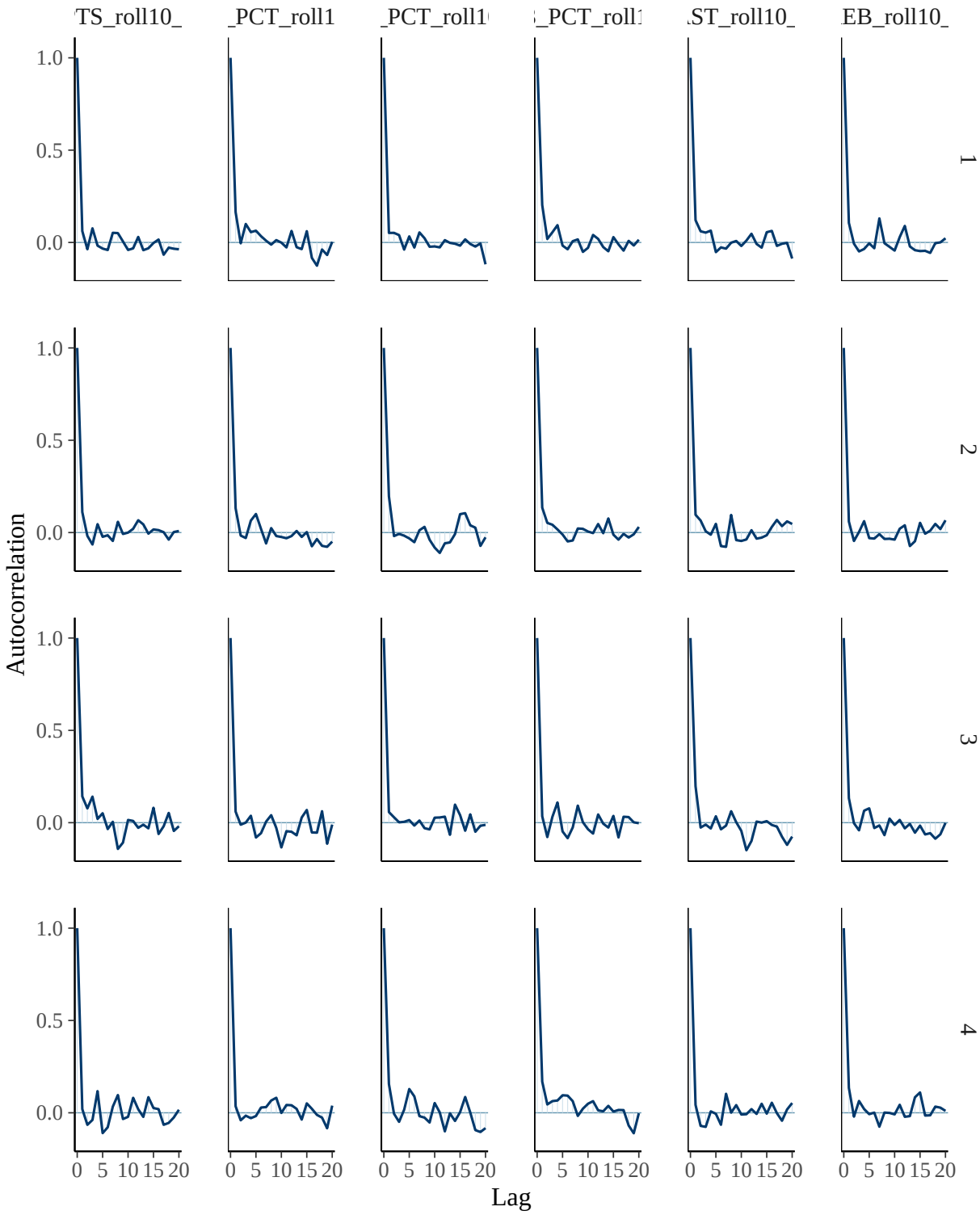

b_PTS_roll10_diff
b_PTS_roll10_diff
b_FG_PCT_roll10_diff
b_FG_PCT_roll10_diff
b_FT_PCT_roll10_diff
b_FT_PCT_roll10_diff
b_FG3_PCT_roll10_diff
b_FG3_PCT_roll10_diff
b_AST_roll10_diff
b_AST_roll10_diff
Chain
1
2
3
4
b_REB_roll10_diff
b_REB_roll10_diff
Chain
1
2
3
4

```r
mcmc_plot(fit, variable = rolling_vars, type = "hist")
```

```
## `stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

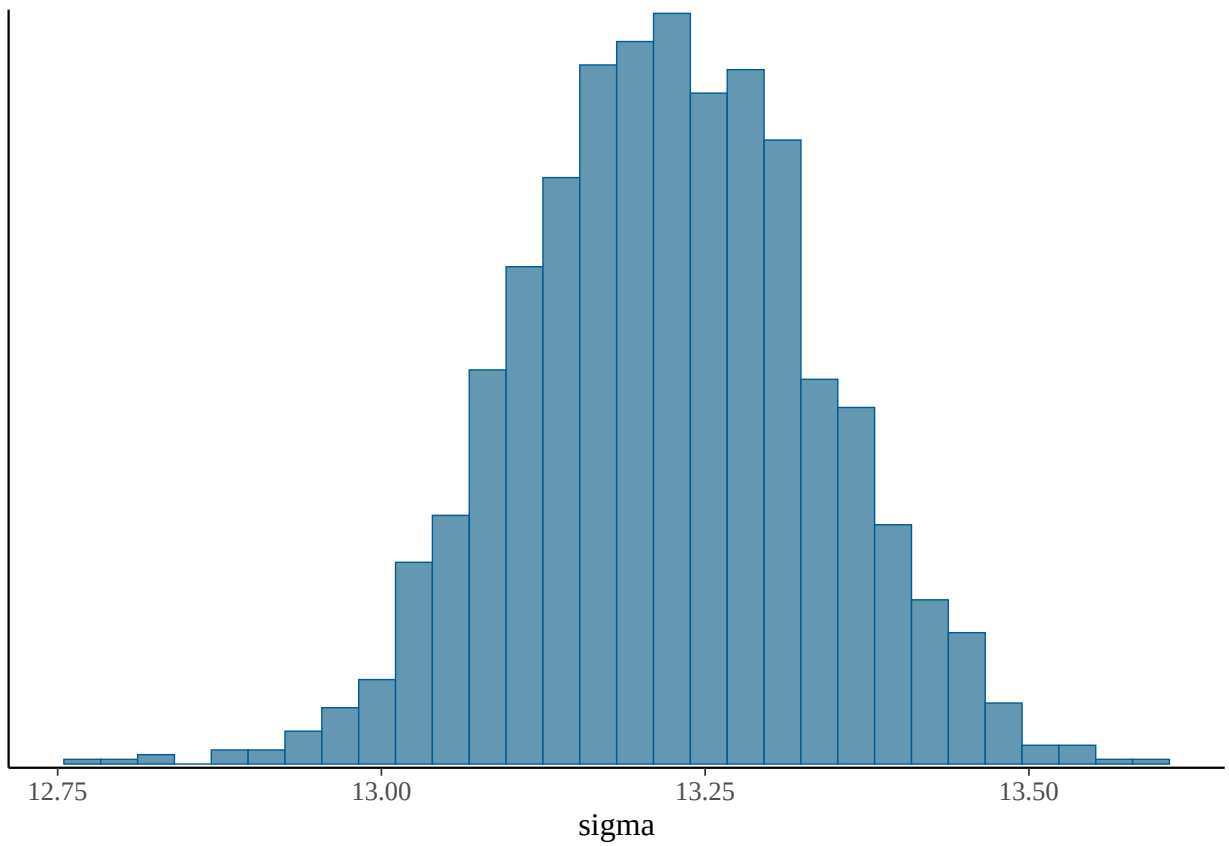


```r
mcmc_plot(fit, variable = rolling_vars, type = "acf")
```




```r
mcmc_plot(fit, variable = "sigma", type = "hist")
```

```
## `stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



evals

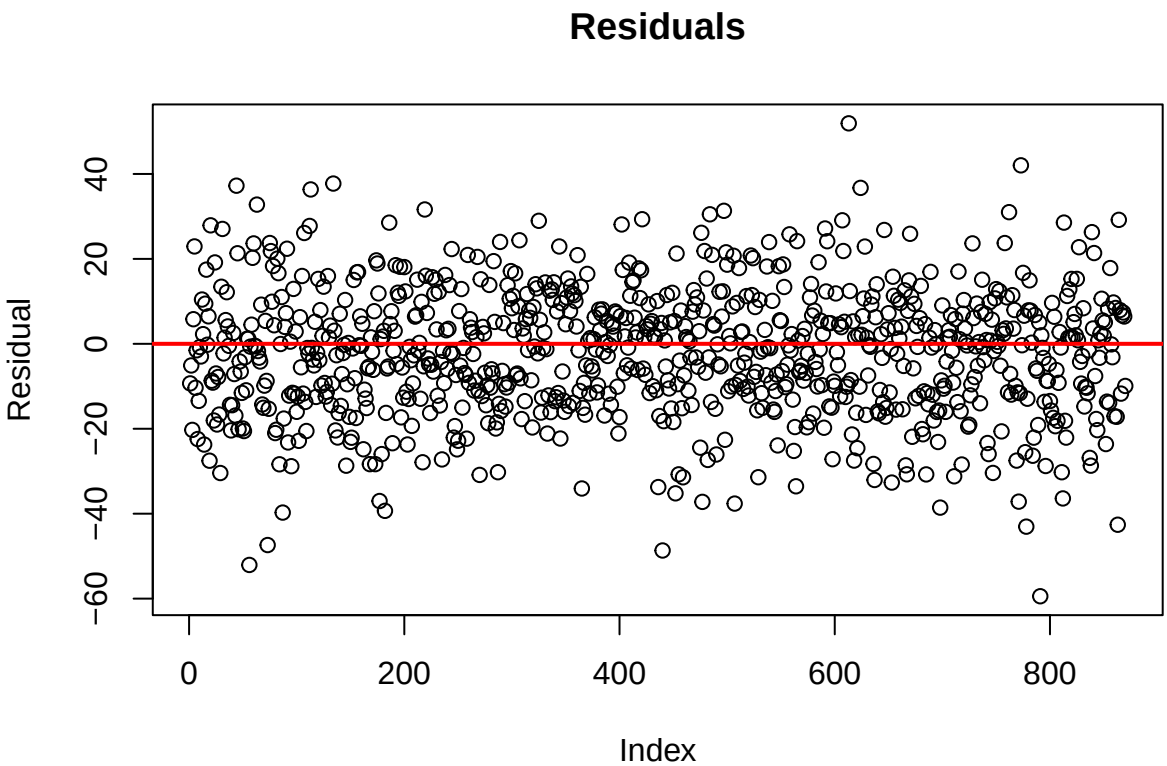
```
bayes_R2(fit)
```

```
##      Estimate   Est.Error       Q2.5      Q97.5
## R2 0.08432919 0.005606297 0.07350774 0.09576567
```

```
post_preds <- posterior_predict(fit, newdata = test_df)

lower <- apply(post_preds, 2, quantile, probs = 0.025)
upper <- apply(post_preds, 2, quantile, probs = 0.975)

coverage <- mean(test_df$POINT_DIFF >= lower & test_df$POINT_DIFF <= upper)
# too much variability, need more informatble priors, so rn inf variants on the weights.
cat(sprintf("95%% posterior predictive interval coverage on test set: %.1f%%\n", coverage*100))
```

```
## 95% posterior predictive interval coverage on test set: 90.9%
```

```
pred_mean <- colMeans(post_preds)

residuals <- test_df$POINT_DIFF - pred_mean

plot(seq_along(residuals), residuals,
     xlab = "Index",
     ylab = "Residual",
     main = "Residuals")
abline(h = 0, col = "red", lwd = 2)
```




```r
mse <- mean((pred_mean - test_df$POINT_DIFF)^2)
cat("MSE (for comparison with the MLP):", mse)
```

```
## MSE (for comparison with the MLP): 227.0225
```